patterns

In [5]:
```
#Right angled trangle

for i in range(1,7):
    print('*  '*i)
```

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
*  *  *  *  *  *
```

In [12]:
```
#inverted right angled trangle
for i in range(7,0,-1):
    print( '* ' *i)
```

```
* * * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*
```

In [ ]:
```
#pyramid pattern
for i in range(1,11,2):
    print('* ' *i)
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * * *
```

In [16]:
```
#inverted pyramid pattern
for i in range(10,0,-2):
    print('* ' *i)
```

```
* * * * * * * * * *
* * * * * * * *
* * * * * *
* * * *
* *
```

In [ ]:
```
#Diamond pattern
for i in range(1,7):
    print('*  ' *i)

for j in range(5,0,-1):
    print('*  '*j)
```

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
*  *  *  *  *  *
*  *  *  *  *
*  *  *  *
*  *  *
*  *
*
```

In [ ]:
```
#square pattern
for i in range(5):
    for j in range(5):
        if i==0 or i==4 or j==0 or j==4:
            print('*',end=' ')

        else:
            print(' ', end=' ')
    print()
```

```
* * * * *
*       *
*       *
*       *
* * * * *
```

In [33]:
```
#full square pattern
for i in range(5):
    print('* * * * * ' )
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

In [ ]:
```
#right angle trangle by using numbers
for i in range(0,7):
    print(' '.join(str(x) for x in range(1, i+1)))
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
```

In [ ]:
```
#inverse right angle trangle by numbers
for i in range(7,0,-1):
    print(' '.join (str(x) for x in range(1, i+1)))
```

```
1 2 3 4 5 6 7
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In [45]:
```
#floyd'd trangle
num=1
for i in range(5):
    for j in range(1,i+1):
        print(num, end=' ')
        num+=1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
```

In [ ]:
```
#hallo right angle trangle
for i in range(1,7):
    for j in range(1,i+1):
        if j==1 or j==i or i==6:
            print('*', end='  ')

        else:
            print('  ', end=' ')
    print()
```

```
*
*  *
*     *
*        *
*           *
*  *  *  *  *  *
```

In [79]:
```
#hallow pyramid pattern
for i in range(1,6):
    for j in range(5 - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        if j==0 or j==2 * i - 2 or i ==5:
            print('*', end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
      *
     *   *
    *       *
   *           *
* * * * * * * * *
```

In [75]:
```
#hallow diamond pattern
n=5
for i in range(1,n+1):
    for j in range(5 - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        if j==0 or j==2 * i - 2:
            print('*', end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
for i in range(n-1,0,-1):
    for j in range(5 - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        if j==0 or j==2 * i - 2 or i ==5:
            print('*', end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
      *
     *   *
    *       *
   *           *
  *               *
 *                   *
  *               *
   *           *
    *       *
     *   *
      *
```

In [82]:
```
#hallow daimaond
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j == 0 or j==2*i-2 or i ==5:
            print(i,end=' ')

        else:
            print(' ', end=' ')
    print()
```

```
      1
     2   2
    3       3
   4           4
  5 5 5 5 5 5 5 5 5
```

In [ ]:
```
n=5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        if j==0 or j== 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ',end=' ')
    print()


    for i in range(n - 1, 0,  -1):
        for j in range(n - i):
            print(' ',end=' ')
```

```
        for j in range(2 * i -  1):
            if j == 0 or j == 2 * i - 2:
                print(i,end=' ')

            else:
                print(' ', end=' ')
        print()
```

```
      1
 4         4
   3     3
     2 2
       1
     2 2
 4         4
   3     3
     2 2
       1
   3     3
 4         4
   3     3
     2 2
       1
 4         4
 4         4
   3     3
     2 2
       1
5             5
  4         4
    3     3
      2 2
        1
```

In [ ]:
```
#hallow daimond numbers

n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j == 0 or j==2*i-2:
            print(i,end=' ')

        else:
            print(' ', end=' ')
    print()


for i in range(n-1,0,-1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j == 0 or j==2*i-2:
            print(i,end=' ')

        else:
            print(' ', end=' ')
    print()
```

```
      1
    2   2
  3       3
4           4
5               5
4           4
  3       3
    2   2
      1
```

In [88]:
```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1                 1
1 2             1 2
1 2 3         1 2 3
1 2 3 4     1 2 3 4
1 2 3 4 5 1 2 3 4 5
```

In [ ]:
```
#butterfly

n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print(j,end=' ')
    print()


for i in range(n-1,0,-1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1                 1
1 2             1 2
1 2 3         1 2 3
1 2 3 4     1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4     1 2 3 4
1 2 3         1 2 3
1 2             1 2
1                 1
```

In [93]:
```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
```

```
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print(j,end=' ')
    print()


for i in range(n,0,-1):
    for j in range(1,i+1):
        print(j,end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1                 1
1 2             1 2
1 2 3         1 2 3
1 2 3 4     1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4     1 2 3 4
1 2 3         1 2 3
1 2             1 2
1                 1
```

In [98]:
```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print('*',end=' ')
    print()


for i in range(n-1,0,-1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print('*',end=' ')
    print()
```

```
*                 *
* *             * *
* * *         * * *
* * * *     * * * *
* * * * * * * * * *
* * * *     * * * *
* * *         * * *
* *             * *
*                 *
```

In [99]:
```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
```

```
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print('*',end=' ')
    print()


for i in range(n,0,-1):
    for j in range(1,i+1):
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(1,i+1):
        print('*',end=' ')
    print()
```

```
*                   *
* *               * *
* * *           * * *
* * * *       * * * *
* * * * * * * * * * *
* * * * * * * * * * *
* * * *       * * * *
* * *           * * *
* *               * *
*                   *
```

In [ ]:
```
n=5
for i in range(1,n+1):

    for j in range(n-i):
        print(' ', end=' ')

    for j in range(2*i-1):
        print('*', end=' ')

    print()
```

```
      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * * *
```

In [102…
```
n=5
for i in range(n,0,-1):

    for j in range(n-i):
        print(' ', end=' ')

    for j in range(2*i-1):
        print('*', end=' ')

    print()
```

```
  * * * * * * * * *
   * * * * * * *
    * * * * *
     * * *
      *
```